

3. Trigonometry

What students need to learn:

Knowledge of secant, cosecant and cotangent and of arcsin, arccos and arctan. Their relationships to sine, cosine and tangent. Understanding of their graphs and appropriate restricted domains.

Knowledge and use of
 $\sec^2 \theta = 1 + \tan^2 \theta$ and
 $\operatorname{cosec}^2 \theta = 1 + \cot^2 \theta$.

Knowledge and use of double angle formulae; use of formulae for $\sin(A \pm B)$, $\cos(A \pm B)$ and $\tan(A \pm B)$ and of expressions for $a \cos \theta + b \sin \theta$ in the equivalent forms of $r \cos(\theta \pm a)$ or $r \sin(\theta \pm a)$.

Angles measured in both degrees and radians.

To include application to half angles. Knowledge of the t ($\tan \frac{1}{2}\theta$) formulae will *not* be required.

Students should be able to solve equations such as $a \cos \theta + b \sin \theta = c$ in a given interval, and to prove simple identities such as $\cos x \cos 2x + \sin x \sin 2x \equiv \cos x$.
